IMD3002 – 3D COMPUTER GRAPHICS

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L15 – Lighting: Indirect Methods

Overview

- This lecture covers the technical properties of various types of lights used in 3D graphics; including Image Based Lighting (IBL)

Image-Based Lighting

- Introduction
  - Image-Based Lighting (IBL) uses a real-world image as an illumination source to light 3D scenes
    - The technique evolved from Reflection Mapping
  - IBL is analogous to Image-Based Modelling (IBM)
    - This technique uses images to derive a 3D scene's geometric structure (Match-Moving can be considered a type of IBM)
  - IBL is also analogous to Image-Based Rendering (IRM) which uses images to produce realistic renderings (either by combining lots of images together, or by using partially rendered 3D scenes)

Image-Based Lighting

- Introduction
  - When used effectively, IBL can produce very realistic rendered images of objects and can be an effective tool for integrating computer graphics objects into real scenes

Image-Based Lighting

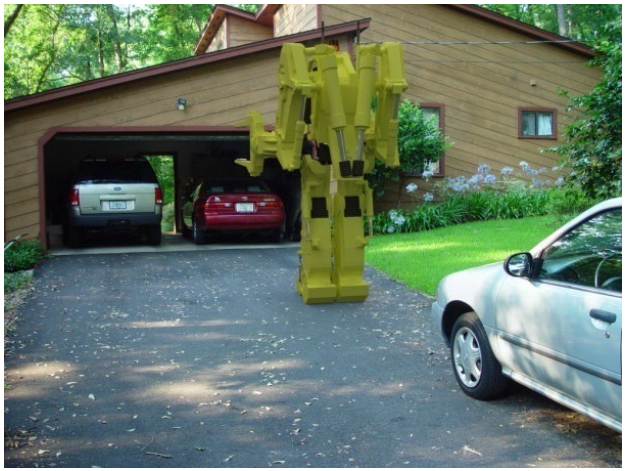
- Introduction
  - **Example:** this image is completely CG with no real objects, the lighting is provided purely by Image-Based Lighting
    - This scene was created in 3DSMax and rendered using global illumination
    - The light was captured in a kitchen, including light from a ceiling fixture, blue sky from the windows, and indirect light from the walls & cabinets

Model: Gary Butcher, **IBL:** Paul Debevec

Image-Based Lighting

- **Example:** models in real environments lit by IBL

This is the infamous G-Man from "Half-Life 2" (the wall is real)



Front loader, similar to the one from "Aliens" whilst the lighting is nice, the surface is poor

Here we have a very nice totally CG environment lit using IBL

Image-Based Lighting

- **Example:** pure CG environments lit by IBL

Image-Based Lighting

- **Example:** comparing real and CG
  - Which objects in this scene are CG?

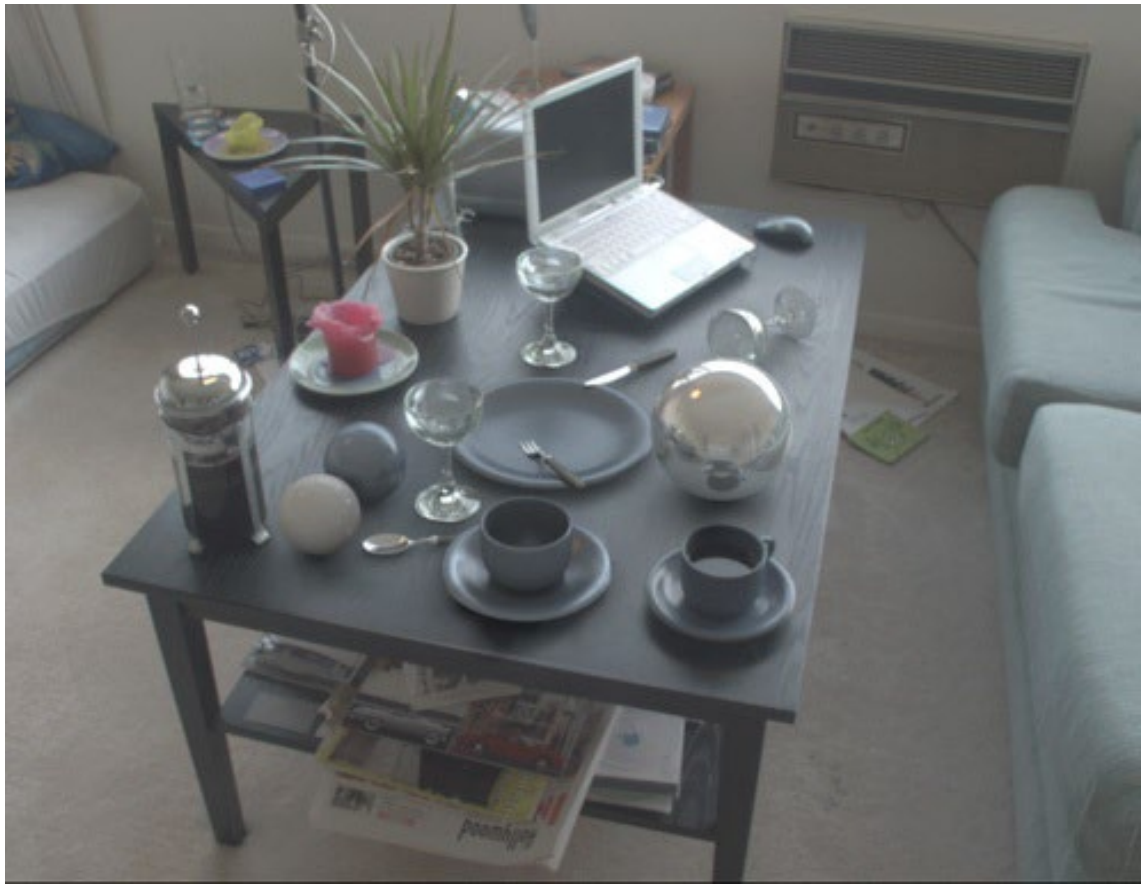


Image-Based Lighting

- **Example:** comparing real and CG
  - Here are the CG objects…

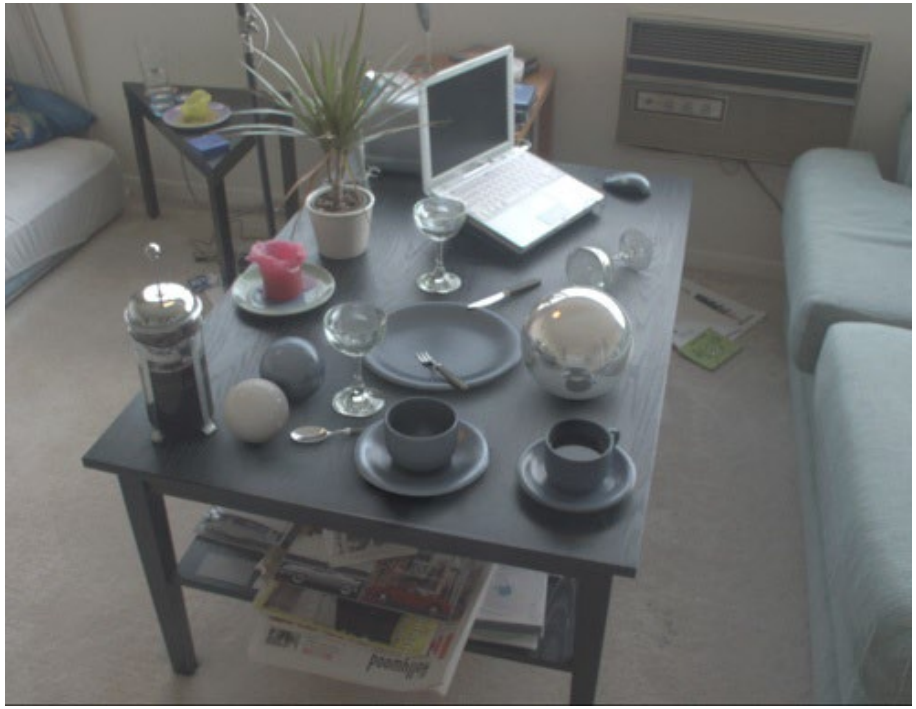


Image-Based Lighting

- Method Overview
  - The major benefit of IBL is that to setup a lighting environment that is relatively realistic is very quick and easy
    - Even if realism is not the goal, IBL can still be used to setup a relatively simple global lighting colour
  - The reason it is so simple is only an image of the light in an environment is required, and therefore any image capturing device (such as a camera) can be used; rendering is performed using relatively simple GI method (such as Final Gather) and therefore the render time is also small

Image-Based Lighting

- Step-By-Step Process
  - There are several steps to producing an IBL-based environment, with a number of options in specific cases:
    - **Capture the lighting** from an environment as an omni-directional, high-dynamic range image (HDRI)
    - **Map the captured image** onto a representation of the environment (usually a sphere)
    - Place the CG scene into the environment
    - **Render the light** from environment using a technique that accounts for the light coming from that environment

Image-Based Lighting

- Capture the Light
  - The first step is to capture the light, in order to do that we need to measure the real-world illumination
    - This is often referred to as a "Light Probe Image"
  - Light Probe Images are usually acquired using a photographic instrument (e.g. a camera), with specific properties
    - The image captured should be omni-directional, so that for every direction in the world there is a corresponding pixel (and thus RGB value) related to that direction
    - The pixel values are linearly proportional to the amount of light in the real world

Image-Based Lighting

- Capture the Light: Omni-Directional
  - Capturing an omni-directional image can be done in several ways (with various benefits):
    - **Mirrored Ball** ~ this is the simplest way; take images of a mirrored ball placed in the scene
      - A mirrored ball actually reflects the entire world around it, not just the hemisphere toward the camera
      - Light rays reflecting off the outer circumference of the ball glance toward the camera from the back half of the environment
    - **Fish-Eye Lens** ~ a good way to cover a large area is to use this kind of lens – this can cover the full field in as few as two images

Image-Based Lighting

- Capture the Light: Omni-Directional
  - Capturing an omni-directional image can be done in several ways (with various benefits), cont'd:
    - **Mosaic** ~ using a regular camera we can take a mosaic of many pictures looking in different directions and then combine them using an image stitching program (e.g. Photoshop)
    - **Panoramic Cameras** ~ we can use a special type of camera (e.g. panoscan, sphereon) which uses a vertical row of image sensors on a rotating camera head to scan across a 360° field-of-view

Image-Based Lighting

- Capture the Light: Linearly Proportional
  - Most digital cameras capture images with pixel values that are not proportional to the scene light
  - The light levels are encoded non-linearly so that they appear more correctly (or pleasingly) on non-linear display devices (such as monitors or TVs)
  - The reason for this is because of the actual range allocated to light levels for typical RGB displays; this provides only a low dynamic range (the ratio between the darkest and brightest region)
    - **Note:** RGB is usually represented either by FF, 255 or 1.0f (depending on the system)

Image-Based Lighting

- Capture the Light: Linearly Proportional
  - The RGB limitation is due to the fact that each pixel value has 256 values for red, 256 values for blue, and 256 values for green
  - Looking at it a different way, the range between the darkest value and the lightest value is only 256
  - All light values have to fit into this range, and therefore depending on the exposure a lot of real light values have to fit into one pixel value
  - In this case, there is a need to use a higher number of bits per channel, such as 16 or 32

Image-Based Lighting

- Capture the Light: Linearly Proportional
  - Example:
    - The 8-bit/channel images (two on the left) don't capture all light values arriving at the camera; however, the High Dynamic Range Image (Right) can store all values

Darker values are saturated

Lighter values are saturated

Image-Based Lighting

- Capture the Light: Linearly Proportional
  - The reason for using 16 or 32-bits/channel, is that it provides a large range to store light information from dark to light
  - With 8-bits per channel, there are only 2^8 (or 256) values for brightness available; therefore for each level of brightness in real life, needs to fit into one of those 256 values
  - With 16-bits/channel, 2^{16} (or 65,536) values per channel are available
  - With 32-bits/channel, 2^{32} (or 4,294,967,296) values per channel are available

Image-Based Lighting

- Capture the Light: Linearly Proportional
  - In most cases the additional information from HDRI cannot be displayed, but the additional data can be used for the Image-Based Lighting
  - Another use of HDRI is by having more light and dark information in an image, especially for each colour, more vivid images can be produced (Using Tone Mapping)

Image-Based Lighting

- Capture the Light: Linearly Proportional
  - HDR images typically use a single-precision floating point number for red, green, and blue, allowing the full range of light from thousandths to billions to be represented
  - HDR images can be stored various file formats, including:
    - **TIFF Floating Pointing** ~ which is simply a floating-point version of the TIFF file format
    - **Portable Floatmap** ~ which is a variant of Jef Postsanzer's Portable Pixmap format

Image-Based Lighting

- Capture the Light: HDRI
  - To produce HDRI using a regular camera, typically involves taking a series of pictures of the scene with varying exposure levels and then using these images to solve for the imaging system's response curve and to form a linear-response composite image covering the entire range of illumination values in the scene
    - **Photoshop (CS3 and above)** has a script called "HDR Merge" constructing an HDRI from several LDR images
    - Alternatively programs such as **HDRShop** (now on version 2.0) handles HDR images

Image-Based Lighting

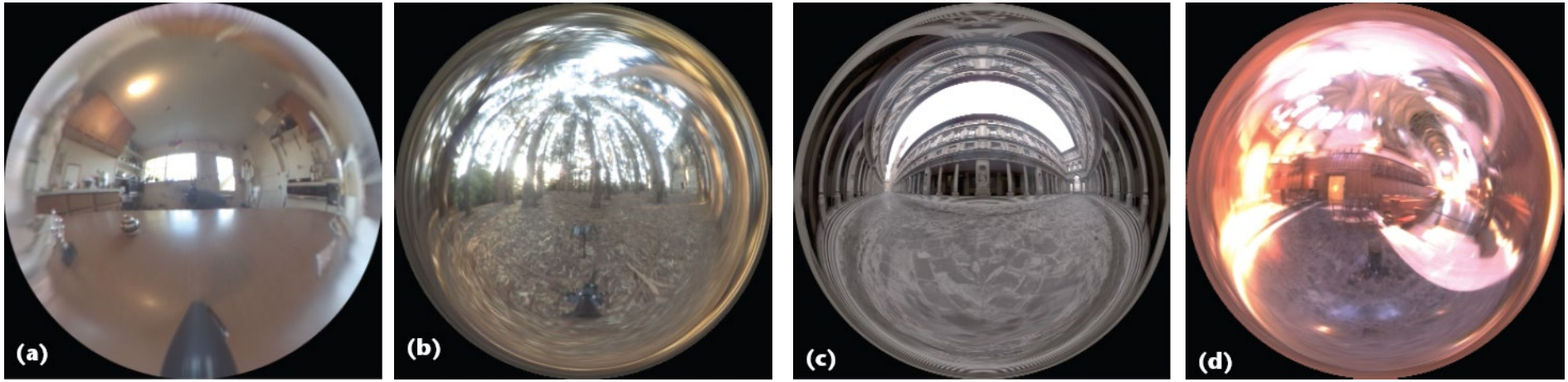
- Capture the Light: HDRI
  - Several other representations that use less storage are available, including Greg Ward's Red-Green-Blue Exponent (RGBE) format
  - The light probe images in the light probe image gallery are in the RGBE format, which lets us easily use them in Ward's Radiance global illumination renderer

Image-Based Lighting

- Capture the Light: HDRI
  - An alternative to capturing the light into an HDRI is to use an already captured image (depending on what is required)
  - There are many light probe images that can be found on the Internet, one of the most popular are those by Paul Debevec (who essentially invented this technique, and won an Academy Award for it)
    - Website: **http://www.debevec.org**
  - There you will find a variety of different images, and other information on IBL;

Image-Based Lighting

- Capture the Light: HDRI
  - **Examples:** various light probe images from Debevec's site



Kitchen	Eucalyptus Grove	Uffizi Gallery, Florence, Italy	Grace Cathedral, San Francisco

Image-Based Lighting

- Capture the Light: HDRI
  - The following are a series of images used in creating a light probe image
  - To acquire these images, a 3” polished ball bearing was placed on top of a tripod
  - A digital camera, with a telephoto zoom lens, was used to take a series of exposures of the ball

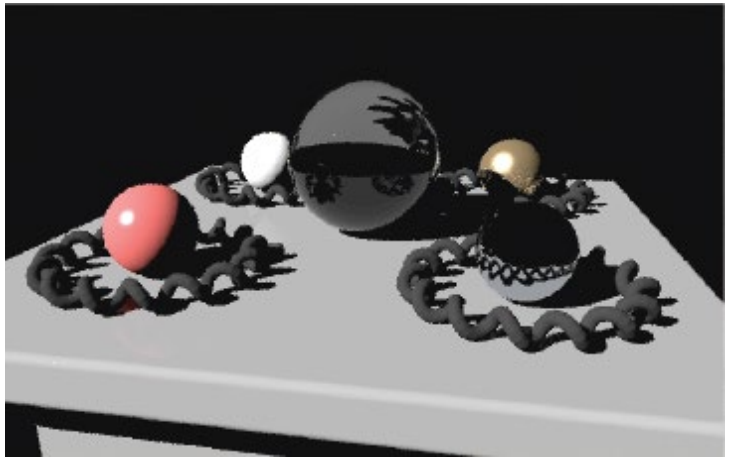
Image-Based Lighting

- Capture the Light: HDRI
  - Care should be taken not to disturb the camera
  - The images were taken at shutter speeds ranging from 1/4th second to 1/10,000th second, allowing the camera to properly image everything from the darkest region to the brightest
  - The images can then be assembled into an HDRI using software (e.g. Photoshop)
    - Note: many cameras are now providing this direct ability on-camera

Image-Based Lighting

- **Examples:** environments rendered using IBL

Original (no IBL)



At the Beach

Kitchen

Forest

Uffizi Gallery

Image-Based Lighting

- Reflectance Field of the Human Face
  - Another use of IBL is to obtain a reflectance map of the human face; this uses a large capture setup of literally hundreds of lights
  - By determining the reflectance properties of a face from every angle, any light can be applied to effectively relight that face (for example from an environment with IBL) and then adjust that real image
  - **Examples:**
    - The following films used this technique: The Matrix 2 (for Agent Smith) and for SpiderMan 2 (for Doc Otto Octavius)

Image-Based Lighting

- Reflectance Field of the Human Face
  - The capture system uses a large array of light positions to capture the light reflecting off an actor's using a camera (from the front) of every position

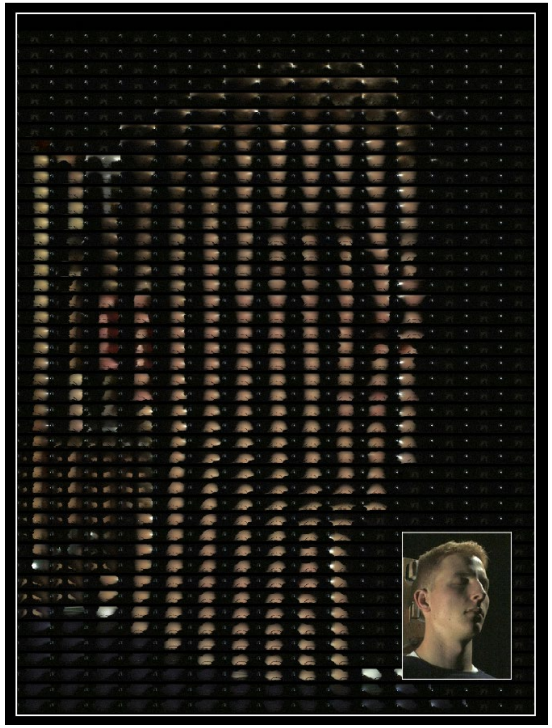
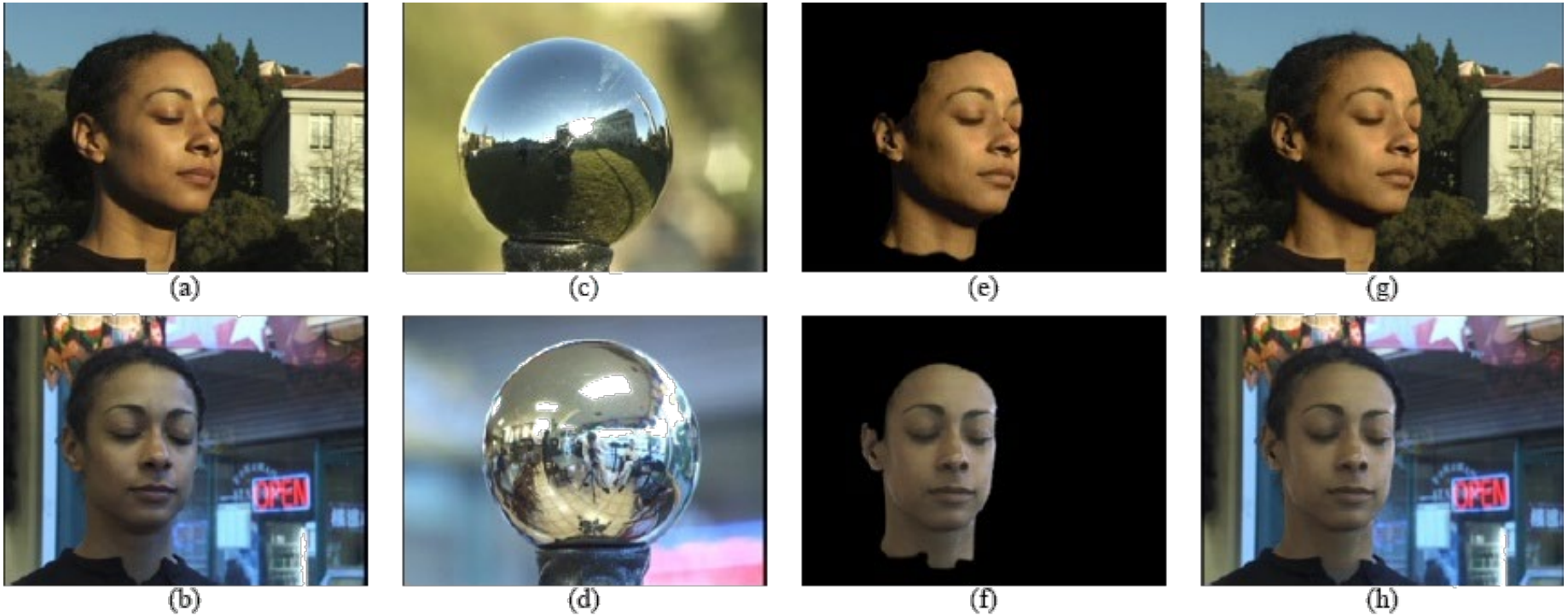


Image-Based Lighting

- Reflectance Field of the Human Face
  - (a, b) real photos (c, d) light probe images (e, f) reflectance field images (g, h) calculated images (c, e) and (d, f) recomposed